La Scuola d'Italia / Dr. Huson / IB Math Name:
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5.2 Classwork: Average Velocity and Displacement

1. A commuter train travels along a straight track from Station A to Station C, passing through Station B.
   - **Stage 1:** It travels East for 15 minutes at a constant speed of 80 km h^{-1}.
   - **Stage 2:** It stops at Station B for 10 minutes.
   - **Stage 3:** It continues East for another 12 km at a constant speed of 48 km h^{-1}.

   (a) Calculate the total displacement from Station A to Station C in kilometers. [2]

   (b) Determine the total time taken for the entire journey in hours. [2]

   (c) Calculate the **average velocity** for the entire journey in km h^{-1}. [2]

2. A drone is programmed to fly along a straight North-South line for a surveying mission.
   - It flies 800 m North in 40 seconds.
   - It hovers in place for 10 seconds to take a photo.
   - It then flies 1200 m South in 60 seconds.

   (a) Calculate the total **distance** traveled by the drone. [1]

   (b) Calculate the final **displacement** of the drone from its starting point. [2]

   (c) Determine the **average speed** of the drone for the total 110-second flight. [2]

   (d) Determine the **average velocity** of the drone for the total flight. [2]

3. A professional cyclist completes a 20 km straight-line time trial.

   - For the first 10 km, she maintains an average velocity of v $\mathrm{km\,h^{-1}}$.
   - For the final 10 km, she increases her effort and maintains an average velocity of $1.5v$ $\mathrm{km\,h^{-1}}$.
   - The total time taken for the entire 20 km trial is 30 minutes.

   (a) Show that the time taken for the first half of the race is $\dfrac{10}{v}$ hours and the second half is $\dfrac{20}{3v}$ hours. [2]

   (b) Calculate the value of v. [3]

   (c) Calculate the cyclist's **average velocity** for the entire 20 km trip. [2]

   (d) Explain why the answer to part (c) is not simply the arithmetic mean of v and $1.5v$. [1]